

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250280882

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

XIAN; Yubin et al.

ELECTRONIC ATOMIZATION DEVICE

Abstract

An electronic atomization device include: a substrate, a heating cavity being formed in the substrate; and an electrode assembly including a first electrode and a second electrode, both the first electrode and the second electrode at least partially extending into the heating cavity. In the heating cavity, an electric arc is controlled to form between the first electrode and the second electrode so as to generate plasma. Negative and positive polarities of the first electrode and the second electrode are switched in a preset cycle.

Inventors: XIAN; Yubin (Shenzhen, CN), LI; Huanxi (Shenzhen, CN), CHEN; Lewen (Shenzhen, CN), ZHOU; Hongming (Shenzhen, CN)

Applicant: SHENZHEN SMOORE TECHNOLOGY LIMITED (Shenzhen, CN)

Family ID: 87171635

Appl. No.: 19/219459

Filed: May 27, 2025

Foreign Application Priority Data

CN 202223171651.6

Nov. 28, 2022

Related U.S. Application Data

parent WO continuation PCT/CN2023/123566 20231009 PENDING child US 19219459

Publication Classification

Int. Cl.: A24F40/46 (20200101); A24F40/42 (20200101); A24F40/57 (20200101); H05B7/101 (20060101); H05H1/48 (20060101)

U.S. Cl.:

Background/Summary

CROSS-REFERENCE TO PRIOR APPLICATION [0001] This application is a continuation of International Patent Application No. PCT/CN2023/123566, filed on Oct. 9, 2023, which claims priority to Chinese Patent Application No. 202223171651.6, filed on Nov. 28, 2022. The entire disclosure of both applications is hereby incorporated by reference herein.

FIELD

[0002] This application relates to the field of atomization technologies, and in particular, to an electronic atomization device.

BACKGROUND

[0003] An aerosol is a colloid dispersion system formed by solid or liquid small particles dispersed and suspended in a gas medium. Since aerosols can be absorbed by a human body through a respiratory system, a new alternative absorption manner is provided for users. For example, an atomization device that can produce the aerosol by baking and heating an aerosol-generating medium of a herbal type or an ointment type is provided, and the atomization device can be used in different fields to deliver an inhalable aerosol to the users, replacing a conventional product form and a conventional absorption manner.

[0004] Generally, the aerosol-generating medium is heated through an electronic atomization device. Some electronic atomization devices use a manner of plasmonic heating. Specifically, the manner of plasmonic heating is to place a high-voltage electrode and a low-voltage electrode in a heating element respectively, with the certain distance between the two electrodes. When a high voltage is loaded on the electrodes, an electric arc and plasma are generated in the gap, and heat generated by the plasma is used for heating.

[0005] However, when a direct current high-voltage power supply is used for discharging, since there is a temperature gradient of the plasma from the high-voltage electrode to the low-voltage electrode in the discharging gap, the heating element has a gradual temperature gradient along the axial direction, which is not conducive to the spatial uniformity of heating. In addition, since a negative electrode is bombarded by a positive ion in the plasma, the temperature of the negative electrode is significantly higher than the temperature of a positive electrode. Although a high-temperature resistant electrode material can slow down the ablation of the electrode to a certain extent, it is still easy for the electrode to be damaged and unable to discharge. Therefore, in a conventional manner of plasmonic heating, the spatial uniformity of heating is not good, and the electrodes are easily damaged.

SUMMARY

[0006] In an embodiment, the present invention provides an electronic atomization device, comprising: a substrate, a heating cavity being formed in the substrate; and an electrode assembly comprising a first electrode and a second electrode, both the first electrode and the second electrode at least partially extending into the heating cavity, wherein, in the heating cavity, an electric arc is controlled to form between the first electrode and the second electrode so as to generate plasma, and wherein negative and positive polarities of the first electrode and the second electrode are switched in a preset cycle.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] Subject matter of the present disclosure will be described in even greater detail below based on the exemplary figures. All features described and/or illustrated herein can be used alone or combined in different combinations. The features and advantages of various embodiments will become apparent by reading the following detailed description with reference to the attached drawings, which illustrate the following:

[0008] FIG. 1 is a schematic structural diagram of an electronic atomization device according to an embodiment of this application;

[0009] FIG. 2 is a schematic diagram of modules of the electronic atomization device shown in FIG. 1;

[0010] FIG. 3 is a schematic structural diagram of an electronic atomization device according to another embodiment of this application; and

[0011] FIG. 4 is a partial schematic structural diagram of the electronic atomization device shown in FIG. 3.

DETAILED DESCRIPTION

[0012] In an embodiment, the present invention provides an electronic atomization device to solve problems of poor spatial uniformity of plasmonic heating and easy damage of electrodes.

[0013] In an embodiment, the present invention provides an electronic atomization device. The electronic atomization device includes: a substrate, a heating cavity being formed in the substrate; and

[0014] an electrode assembly, including a first electrode and a second electrode, both the first electrode and the second electrode at least partially extending into the heating cavity, where [0015] in the heating cavity, an electric arc is controlled to form between the first electrode and the second electrode, plasma is generated, and negative and positive polarities of the first electrode and the second electrode are switched in a preset cycle.

[0016] In the foregoing electronic atomization device, in the heating cavity, the electric arc can be controlled to form between the first electrode and the second electrode, and the plasma can be generated, so that the substrate is heated through the plasma generated by discharging between the first electrode and the second electrode, thereby heating and atomizing an aerosol-generating medium disposed on the substrate to form an aerosol for users to inhale.

[0017] In addition, the negative and positive polarities of the first electrode and the second electrode are switched in the preset cycle. In other words, in a period of time, the first electrode is a negative electrode and the second electrode is a positive electrode, and in another period of time, reverse switching is performed, the first electrode is the positive electrode, and the second electrode is the negative electrode. Such reverse switching enables the first electrode and the second electrode to be alternately used as the negative electrode, thereby preventing one of the first electrode and the second electrode from being ablated by the high temperature due to being the negative electrode for a long time and preventing the electrodes from being damaged and unable to discharge. In addition, the first electrode and the second electrode alternately have the negative and positive polarities, so that the negative electrode with the high temperature alternately appears on the first electrode and the second electrode. In this way, the substrate has the high temperature at one end for a period of time and the high temperature at the other end for another period of time. After switching for several times, the temperature of the entire substrate is more uniformly in space, and the substrate can more uniformly heat and atomize the aerosol-generating medium. In this way, the uniformity of a heating space heated by the plasma is better, and the electrodes are not easily damaged.

[0018] In an embodiment, the electronic atomization device includes a power supply assembly and a control assembly, where the control assembly is electrically connected between the power supply assembly and the electrode assembly, the power supply assembly is configured to provide a high-

voltage alternating current, and the control assembly is configured to output a high-voltage direct current to the electrode assembly and switch the negative and positive polarities of the first electrode and the second electrode in the preset cycle.

[0019] In an embodiment, the power supply assembly includes a power supply module and a transformer, and the transformer is connected to the power supply module and is configured to output the high-voltage alternating current; and

[0020] the control assembly includes a rectifier and a switch control, the rectifier is electrically connected between the transformer and the switch control and is configured to integrate the high-voltage alternating current into a high-voltage direct current that flows to the switch control, both the first electrode and the second electrode are electrically connected to the switch control, and the switch control is configured to reversely switch the negative and positive polarities of the first electrode and the second electrode in the preset cycle.

[0021] In an embodiment, the substrate is constructed as a center heating structure inserted into an aerosol-generating medium, or the substrate is constructed as a peripheral heating structure surrounding the aerosol-generating medium.

[0022] In an embodiment, the substrate includes a tubular body, the tubular body includes the heating cavity, both the first electrode and the second electrode at least partially extend into the heating cavity, and end portions of the first electrode and the second electrode are disposed at an interval along the axial direction of the tubular body.

[0023] In an embodiment, the tubular body includes the top end and the bottom end that are opposite to each other along the axial direction of the tubular body, one of the first electrode and the second electrode whose end portion along the axial direction is located at the top end is a negative electrode in the initial polarity, and the other of the first electrode and the second electrode whose end portion along the axial direction is located at the bottom end is a positive electrode in the initial polarity.

[0024] In an embodiment, the tubular body includes an inner tubular body and an outer tubular body that is sleeved outside the inner tubular body at an interval, the heating cavity includes a first sub-cavity and a second sub-cavity that are in communication with each other, the first sub-cavity is disposed throughout the inner tubular body along the axial direction of the inner tubular body, and the second sub-cavity is defined between the top portion of the inner tubular body and the outer tubular body; and

[0025] one of the first electrode and the second electrode at least partially extends into the first sub-cavity, and the other of the first electrode and the second electrode at least partially extends into the second sub-cavity.

[0026] In an embodiment, one end of the first electrode extends into the first sub-cavity, the other end of the first electrode is located outside the first sub-cavity, the second electrode includes a body portion and a pin portion, the body portion is disposed in the second sub-cavity and is disposed opposite to the first electrode at an interval along the axial direction of the inner tubular body, and the pin portion is connected to the body portion and extends to the outside through the gap between the inner tubular body and the outer tubular body.

[0027] In an embodiment, the substrate includes a shell and an inner pot sleeved on the shell, the inner pot includes an accommodating cavity, the heating cavity is defined between the inner pot and the shell, and both the first electrode and the second electrode are mounted on the shell.

[0028] In an embodiment, one of the first electrode and the second electrode is disposed around the periphery of the inner pot, and the other of the first electrode and the second electrode is located at the center of the bottom portion of the inner pot; and

[0029] one of the first electrode and the second electrode that is at the center of the bottom portion of the inner pot is the negative electrode in the initial polarity, and the other of the first electrode and the second electrode that is around the periphery of the inner pot is the positive electrode in the initial polarity.

[0030] In an embodiment, the inner pot is detachably disposed on the shell.

[0031] Descriptions of referential numerals: **100**: Atomizer; **10**: Substrate; **11**: Heating cavity; **112**: First sub-cavity; **114**: Second sub-cavity; **13**: Accommodating cavity; **14**: Tubular body; **141**: Top end; **143**: Bottom end; **145**: Inner tubular body; **147**: Outer tubular body; **16**: Shell; **18**: Inner pot; **30**: Electrode assembly; **32**: First electrode; **34**: Second electrode; **341**: Body portion; **343**: Pin portion; **50**: Power supply assembly; **52**: Power supply module; **54**: Transformer; **70**: Control assembly; **72**: Rectifier; **74**: Switch control; and **80**: Magnetic ring.

[0032] To make the foregoing objects, features, and advantages of this application clearer and easier to understand, the following further describes specific implementations of this application with reference to the accompanying drawings. In the following descriptions, many specific details are described to fully understand this application. However, this application can be implemented in many other manners different from those described herein. A person skilled in the art can make similar improvements without departing from the connotation of this application. Therefore, this application is not limited to the specific embodiments disclosed below.

[0033] In the descriptions of this application, it is to be understood that orientation or position relationships indicated by terms such as “center”, “longitudinal”, “transverse”, “length”, “width”, “thickness”, “upper”, “lower”, “front”, “rear”, “left”, “right”, “vertical”, “horizontal”, “top”, “bottom”, “inner”, “outer”, “clockwise”, “anticlockwise”, “axial”, “radial”, and “circumferential” are based on orientation or position relationships shown in the accompanying drawings, and are used only for ease of describing this application and simplifying the descriptions description, rather than indicating or implying that the mentioned device or component have to have a particular orientation or have to be constructed and operated in a particular orientation. Therefore, such terms cannot be construed as a limitation to this application.

[0034] In addition, terms “first” and “second” are merely for description, and cannot be understood as indicating or implying relative importance or implying a quantity of technical features indicated. Therefore, features defined by “first” and “second” may explicitly indicate or implicitly include at least one of such features. In the descriptions of this application, “a plurality of” means at least two, such as two or three, unless otherwise explicitly and specifically defined.

[0035] In this application, unless otherwise explicitly specified or defined, terms such as “mount”, “connected”, “connection”, and “fix” should be understood in a broad sense. For example, the connection may be a fixed connection, a detachable connection, or an integral connection; or the connection may be a mechanical connection, or an electrical connection; or the connection may be a direct connection, an indirect connection through an intermediary, or internal communication between two elements or a mutual action relationship between two elements, unless otherwise specified explicitly. A person of ordinary skill in the art may understand specific meanings of the foregoing terms in this application based on specific situations.

[0036] In this application, unless otherwise explicitly specified and defined, that a first feature is “above” or “below” a second feature may be that the first feature directly contacts with the second feature, or the first feature indirectly contacts with the second feature through an intermediary. In addition, that the first feature is “above”, “over”, or “on” the second feature may be that the first feature is directly above or obliquely above the second feature, or may merely indicate that the first feature is at a higher horizontal position than the second feature. That the first feature is “under”, “below”, and “on the bottom of” the second feature may be that the first feature is directly below or obliquely below the second feature, or may merely indicate that the first feature is at a lower horizontal position than the second feature.

[0037] It should be noted that, when an element is referred to as being “fixed to” or “disposed on” another element, the element may be directly on the another element, or there may be an intervening element. When an element is considered to be “connected to” another element, the element may be directly connected to the another element, or there may be an intervening element. The terms “vertical”, “horizontal”, “upper”, “lower”, “left”, “right”, and similar expressions used

herein are merely intended for description, and do not indicate a unique implementation.

[0038] Referring to FIG. 1, in an embodiment of this application, an electronic atomization device **100** is provided and includes a substrate **10** and an electrode assembly **30**. A heating cavity **11** is formed in the substrate **10**, the electrode assembly **30** includes a first electrode **32** and a second electrode **34**, and both the first electrode **32** and the second electrode **34** at least partially extend into the heating cavity **11**. In the heating cavity **11**, an electric arc can be controlled to form between the first electrode **32** and the second electrode **34**, and plasma can be generated, so that the substrate **10** is heated through the plasma generated by discharging between the first electrode **32** and the second electrode **34**, thereby heating and atomizing an aerosol-generating medium disposed on the substrate **10** to form an aerosol for users to inhale.

[0039] In addition, the negative and positive polarities of the first electrode **32** and the second electrode **34** are switched in a preset cycle. In other words, in a period of time, the first electrode **32** is a negative electrode and the second electrode **34** is a positive electrode, and in another period of time, reverse switching is performed, the first electrode **32** is the positive electrode, and the second electrode **34** is the negative electrode. Such reverse switching enables the first electrode **32** and the second electrode **34** to be alternately used as the negative electrode, thereby preventing one of the first electrode **32** and the second electrode **34** from being ablated by the high temperature due to being the negative electrode for a long time and preventing the electrodes from being damaged and unable to discharge. In addition, the first electrode **32** and the second electrode **34** alternately have the negative and positive polarities, so that the negative electrode with the high temperature alternately appears on the first electrode **32** and the second electrode **34**. In this way, the substrate **10** has the high temperature at one end for a period of time and the high temperature at the other end for another period of time. After switching for several times, the temperature of the entire substrate **10** is more uniformly in space, and the substrate **10** can more uniformly heat and atomize the aerosol-generating medium. In this way, the uniformity of a heating space heated by the plasma is better, and the electrodes are not easily damaged. Referring to FIG. 2, further, the electronic atomization device **100** includes a power supply assembly **50** and a control assembly **70**, the control assembly **70** is electrically connected between the power supply assembly **50** and the electrode assembly **30**, the power supply assembly **50** is configured to provide a high-voltage alternating current, and the control assembly **70** is configured to output a high-voltage direct current to the electrode assembly **30** and switch the negative and positive polarities of the first electrode **32** and the second electrode **34** in the preset cycle. In this way, the control assembly **70** is electrically connected between the power supply assembly **50** and the electrode assembly **30**, which can convert the high-voltage alternating current into the high-voltage direct current that flows to the electrode assembly **30** through the control assembly **70**, and can further switch the negative and positive polarities in the preset cycle through the control assembly **70**, so that the polarities of the first electrode **32** and the second electrode **34** change in cycle. It should be noted that, a switching duration of a cyclic switching may be set to a same duration, or may be set to different durations according to requirements, to achieve a regulated movement of a high-temperature region between the first electrode **32** and the second electrode **34** by adjusting a proportion of a duration in which the first electrode **32** or the second electrode **34** is used as the negative electrode.

[0040] Specifically, the power supply assembly **50** includes a power supply module **52** and a transformer **54**. The transformer **54** is connected to the power supply module **52** and is configured to output the high-voltage alternating current, to step up a current by using the transformer **54**. The control assembly **70** includes a rectifier **72** and a switch control **74**, the rectifier **72** is electrically connected between the transformer **54** and the switch control **74** and is configured to integrate the high-voltage alternating current into a high-voltage direct current that flows to the switch control **74**, both the first electrode **32** and the second electrode **34** are electrically connected to the switch control **74**, and the switch control **74** is configured to reversely switch the negative and positive polarities of the first electrode **32** and the second electrode **34** in the preset cycle. In this way, the

high-voltage alternating current outputted by the transformer **54** is converted into the high-voltage direct current through the rectifier **72**, then the high-voltage direct current is outputted to the electrode assembly **30** through the switch control **74**, and the polarities of the first electrode **32** and the second electrode **34** are switched in cycle through the switch control **74**, so that the first electrode **32** and the second electrode **34** are intermittently used as the negative electrode, to prevent the first electrode **32** and the second electrode **34** from being used as the negative electrode and being burned at the high temperature for a long time. In addition, the high-temperature region corresponding to the negative electrode on the substrate **10** is constantly switched, and heating on the entire substrate **10** is more uniform.

[0041] Referring to FIG. **1**, in some embodiments, the substrate **10** is constructed as a center heating structure inserted into the aerosol-generating medium. When the plasma is generated in the heating cavity **11** inside the substrate **10**, the substrate **10** is heated, and the substrate **10** is inserted into the aerosol-generating medium, to transfer heat from the inside of the aerosol-generating medium, to heat and atomize the aerosol-generating medium.

[0042] Further, the substrate **10** includes a tubular body **14**, the tubular body **14** includes the heating cavity **11**, both the first electrode **32** and the second electrode **34** at least partially extend into the heating cavity **11**, and end portions of the first electrode **32** and the second electrode **34** are disposed at an interval along the axial direction of the tubular body **14**. In this way, the electric arc is formed inside the tubular body **14** along the axial direction of the tubular body **14**, and the tubular body **14** is heated along the axial direction of the tubular body **14**. During use, the aerosol-generating medium is inserted into the tubular body **14**, so that the aerosol-generating medium can be heated from the inside to the outside through the tubular body **14**.

[0043] Specifically, the tubular body **14** includes the top end **141** and the bottom end **143** that are opposite to each other along the axial direction of the tubular body **14**, one of the first electrode **32** and the second electrode **34** whose end portion along the axial direction is located at the top end **141** is a negative electrode in the initial polarity, and the other of the first electrode **32** and the second electrode **34** whose end portion along the axial direction is located at the bottom end **143** is a positive electrode in the initial polarity. In this way, the negative electrode with the higher temperature is located at the top end **141**, so that the high temperature region is closer to the top end **141**, thereby improving an initial smoke output speed.

[0044] Specifically, in this embodiment, the tubular body **14** includes an inner tubular body **145** and an outer tubular body **147** that is sleeved outside the inner tubular body **145** at an interval, the heating cavity **11** includes a first sub-cavity **112** and a second sub-cavity **114** that are in communication with each other, the first sub-cavity **112** is disposed throughout the inner tubular body **145** along the axial direction of the inner tubular body **145**, and the second sub-cavity **114** is defined between the top portion of the inner tubular body **145** and the outer tubular body **147**. One of the first electrode **32** and the second electrode **34** at least partially extends into the first sub-cavity **112**, and the other of the first electrode **32** and the second electrode **34** at least partially extends into the second sub-cavity **114**. In this way, after the inner tubular body **145** and the outer tubular body **147** are sleeved to form the heating cavity **11**, and the first electrode **32** and the second electrode **34** are disposed at an interval along the axial direction of the inner tubular body **145**, the plasma can be controlled to at least form in the first sub-cavity **112** along the axial direction of the inner tubular body **145**, to heat the tubular body **14** along the axial direction.

[0045] Further, one end of the first electrode **32** extends into the first sub-cavity **112**, and the other end of the first electrode **32** is located outside the first sub-cavity **112** for circuit connection. The second electrode **34** includes a body portion **341** and a pin portion **343**. The body portion **341** is disposed in the second sub-cavity **114**, and is disposed opposite to the first electrode **32** at an interval along the axial direction of the inner tubular body **145**. The pin portion **343** is connected to the body portion **341** and extends to the outside through the gap between the inner tubular body **145** and the outer tubular body **147**, so that the second electrode **34** is connected to the circuit. In

addition, the electric arc is generated in the first sub-cavity **112** along the axial direction through the first electrode **32** and the body portion **341** of the second electrode **34**, to heat the tubular body **14**.

[0046] Referring to FIG. **3** and FIG. **4**, in some other embodiments, the substrate **10** is constructed as a peripheral heating structure surrounding the aerosol-generating medium. In other words, the substrate **10** surrounds the aerosol-generating medium. The substrate **10** is heated when the plasma is generated in the heating cavity **11**, and the substrate **10** heats, from the outside to the inside, the aerosol-generating medium surrounded by the substrate **10**, thereby atomizing the aerosol-generating medium to form the aerosol for the users to inhale.

[0047] Further, the substrate **10** includes a shell **16** and an inner pot **18** sleeved on the shell **16**. The inner pot **18** includes an accommodating cavity **13**. The heating cavity **11** is defined between the inner pot **18** and the shell **16**. Both the first electrode **32** and the second electrode **34** are mounted on the shell **16**, and the first electrode **32** and the second electrode **34** can at least partially extend into the heating cavity **11** between the inner pot **18** and the shell **16**, to heat the inner pot **18** after the electric arc and the plasma are generated in the heating cavity **11** by discharging, thereby heating and atomizing the aerosol-generating medium in the accommodating cavity **13** of the inner pot **18**.

[0048] Specifically, one of the first electrode **32** and the second electrode **34** is disposed around the periphery of the inner pot **18**, and the other of the first electrode **32** and the second electrode **34** is located at the center of the bottom portion of the inner pot **18**. In this way, the electric arc formed between the first electrode **32** and the second electrode **34** may rotate around the entire periphery of the inner pot **18** based on driving of a magnetic field, to heat the entire periphery of the inner pot **18**, so that a heating region is more uniform. Optionally, the second electrode **34** includes a discharging ring sleeved on the periphery of the inner pot **18**. The electronic atomization device **100** further includes a magnetic ring **80** disposed on the shell **16** and sleeved on the periphery of the discharging ring, so that the arc is driven to rotate around the periphery of the discharging ring and in the heating cavity **11**, to uniformly heat the inner pot **18** from the entire periphery of the inner pot **18** by using the magnetic ring **80**.

[0049] In addition, one of the first electrode **32** and the second electrode **34** that is at the center of the bottom portion of the inner pot **18** is the negative electrode in the initial polarity, and the other of the first electrode **32** and the second electrode **34** that is around the periphery of the inner pot **18** is the positive electrode in the initial polarity. In this way, in an initial state, the temperature at the center of the bottom portion of the inner pot **18** is high, and the temperature at the bottom portion of the inner pot **18** is quickly concentrated, thereby shortening a smoke output duration in an initial stage and increasing the smoke output speed in the initial stage of atomization. Subsequently, after a requirement for rapid smoke output is satisfied, the polarities of the first electrode **32** and the second electrode **34** are reversely switched, so that one of the first electrode **32** and the second electrode **34** at the periphery of the inner pot **18** is the negative electrode, and the other of the first electrode **32** and the second electrode **34** at the center of the bottom portion of the inner pot **18** is the positive electrode. In this way, a temperature field at the periphery of the inner pot **18** is uniform, and the aerosol-generating medium can be uniformly heated.

[0050] It may be understood that, the polarities of the first electrode **32** and the second electrode **34** may be switched for a single time, or may be switched in cycle for several times. This is not limited herein.

[0051] Optionally, the inner pot **18** is detachably disposed on the shell **16**. When the aerosol-generating medium is heated and atomized, the aerosol-generating medium is disposed in the accommodating cavity **13** of the inner pot **18**, and the inner pot **18** is mounted on the shell **16**. The heating cavity **11** is defined between the inner pot **18** and the shell **16**. The electrode assembly **30** discharges in the heating cavity **11** to generate the electric arc and the plasma, to heat the inner pot **18** and the aerosol-generating medium in the inner pot **18** through the heat of the plasma. After the

electronic atomization device **100** is used for a period of time, the inner pot **18** may be conveniently detached from the shell **16** for cleaning, to prevent, after the inner pot **18** is used for a long time, a residual in the inner pot **18** from affecting the atomization taste, thereby ensuring the atomization taste.

[0052] The technical features in the foregoing embodiments may be randomly combined. For concise descriptions, not all possible combinations of the technical features in the foregoing embodiments are described. However, provided that combinations of the technical features do not conflict with each other, the combinations of the technical features are considered as falling within the scope described in this specification.

[0053] While the invention has been illustrated and described in detail in the drawings and foregoing description, such illustration and description are to be considered illustrative or exemplary and not restrictive. It will be understood that changes and modifications may be made by those of ordinary skill within the scope of the following claims. In particular, the present invention covers further embodiments with any combination of features from different embodiments described above and below. Additionally, statements made herein characterizing the invention refer to an embodiment of the invention and not necessarily all embodiments.

[0054] The terms used in the claims should be construed to have the broadest reasonable interpretation consistent with the foregoing description. For example, the use of the article “a” or “the” in introducing an element should not be interpreted as being exclusive of a plurality of elements. Likewise, the recitation of “or” should be interpreted as being inclusive, such that the recitation of “A or B” is not exclusive of “A and B,” unless it is clear from the context or the foregoing description that only one of A and B is intended. Further, the recitation of “at least one of A, B and C” should be interpreted as one or more of a group of elements consisting of A, B and C, and should not be interpreted as requiring at least one of each of the listed elements A, B and C, regardless of whether A, B and C are related as categories or otherwise. Moreover, the recitation of “A, B and/or C” or “at least one of A, B or C” should be interpreted as including any singular entity from the listed elements, e.g., A, any subset from the listed elements, e.g., A and B, or the entire list of elements A, B and C.

Claims

1. An electronic atomization device, comprising: a substrate, a heating cavity being formed in the substrate; and an electrode assembly comprising a first electrode and a second electrode, both the first electrode and the second electrode at least partially extending into the heating cavity, wherein, in the heating cavity, an electric arc is controlled to form between the first electrode and the second electrode so as to generate plasma, and wherein negative and positive polarities of the first electrode and the second electrode are switched in a preset cycle.
2. The electronic atomization device of claim 1, further comprising: a power supply assembly; and a control assembly, wherein the control assembly is electrically connected between the power supply assembly and the electrode assembly, wherein the power supply assembly is configured to provide a high-voltage alternating current, and wherein the control assembly is configured to output a high-voltage direct current to the electrode assembly and switch the negative and positive polarities of the first electrode and the second electrode in the preset cycle.
3. The electronic atomization device of claim 2, wherein the power supply assembly comprises a power supply module and a transformer, the transformer is connected to the power supply module, the transformer being configured to output the high-voltage alternating current, wherein the control assembly comprises a rectifier and a switch control, the rectifier is electrically connected between the transformer and the switch control, the rectifier being configured to integrate the high-voltage alternating current into a high-voltage direct current that flows to the switch control, wherein both the first electrode and the second electrode are electrically connected to the switch control, and

wherein the switch control is configured to reversely switch the negative and positive polarities of the first electrode and the second electrode in the preset cycle.

4. The electronic atomization device of claim 1, wherein the substrate comprises a center heating structure inserted into an aerosol-generating medium, or wherein the substrate comprises a peripheral heating structure surrounding the aerosol-generating medium.

5. The electronic atomization device of claim 4, wherein the substrate comprises a tubular body, the tubular body comprising the heating cavity, wherein both the first electrode and the second electrode at least partially extend into the heating cavity, and wherein end portions of the first electrode and the second electrode are disposed at an interval along an axial direction of the tubular body.

6. The electronic atomization device of claim 5, wherein the tubular body comprises a top end and a bottom end that are opposite to each other along the axial direction of the tubular body, wherein one of the first electrode and the second electrode whose end portion along the axial direction is located at the top end comprises a negative electrode in the initial polarity, and wherein an other of the first electrode and the second electrode whose end portion along the axial direction is located at the bottom end comprises a positive electrode in the initial polarity.

7. The electronic atomization device of claim 5, wherein the tubular body comprises an inner tubular body and an outer tubular body that is sleeved outside the inner tubular body at an interval, wherein the heating cavity comprises a first sub-cavity and a second sub-cavity that are in communication with each other, the first sub-cavity being disposed throughout the inner tubular body along the axial direction of the inner tubular body, and the second sub-cavity being defined between the top portion of the inner tubular body and the outer tubular body, wherein one of the first electrode and the second electrode at least partially extends into the first sub-cavity, and wherein an other of the first electrode and the second electrode at least partially extends into the second sub-cavity.

8. The electronic atomization device of claim 7, wherein one end of the first electrode extends into the first sub-cavity, wherein an other end of the first electrode is located outside the first sub-cavity, wherein the second electrode comprises a body portion and a pin portion, wherein the body portion is disposed in the second sub-cavity and is disposed opposite to the first electrode at an interval along the axial direction of the inner tubular body, and wherein the pin portion is connected to the body portion and extends to an outside through a gap between the inner tubular body and the outer tubular body.

9. The electronic atomization device of claim 1, wherein the substrate comprises a shell and an inner pot sleeved on the shell, wherein the inner pot comprises an accommodating cavity, wherein the heating cavity is defined between the inner pot and the shell, and wherein both the first electrode and the second electrode are mounted on the shell.

10. The electronic atomization device of claim 9, wherein one of the first electrode and the second electrode is disposed around a periphery of the inner pot, and wherein an other of the first electrode and the second electrode is located at a center of a bottom portion of the inner pot, and wherein one of the first electrode and the second electrode that is at the center of the bottom portion of the inner pot comprises the negative electrode in the initial polarity, and wherein the other of the first electrode and the second electrode that is around the periphery of the inner pot comprises the positive electrode in the initial polarity.

11. The electronic atomization device of claim 10, wherein the inner pot is detachably disposed on the shell.
